



PROJECT REPORT

CSC435 OBJECT-ORIENTED PROGRAMMING

PROJECT:

ELITE TUITION CENTRE: TUITION REGISTRATION SYSTEM

TEAM MEMBERS:

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CLASS:

M3CDCS2303B

PROVIDED FOR:

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TASK FOR EACH MEMBER

NAME OF THE MEMBER	MEMBER'S PICTURE	POSITION	TASK
NURUL NAJWA		PROJECT MANAGER	- Oversees the entire project, ensuring all components are aligned and integrated effectively. - Manages the timeline, resources, and tasks to ensure deadlines are met. - Coordinates between all team members and resolves conflicts or issues. - Ensures communication with stakeholders and provides regular updates on project progress.
NURUL AIN SAFIYYAH		BACKEND DEVELOPER	- Develops the logic and processing behind the application features (excluding the GUI). - Implements the logic to calculate tuition fees and generate receipts. - Manages the connection between the frontend interface and the data stored in text files. - Ensures accurate handling of data related to student and teacher selections.
NOR NAJWA ALI'IN BINTI KAIZA		FILE-HANDLING DEVELOPER	- Handles the data storage using text files, ensuring that student, teacher, and tuition package information is saved and retrieved properly. - Develops functions to manage data input/output between the application and text files. - Ensures that the file management is efficient and maintains data integrity without data loss or errors. - Implements proper data validation before writing to or reading from the text files
SITI RAUDAH BINTI ABU BAKAR		FRONTEND DEVELOPER	- Develops the graphical user interface (GUI) using BlueJ. - Designs and implements the layout of the application, including buttons, forms, and display areas. - Integrates the logic for user interactions, such as subject and teacher selection, and displaying student/teacher information. - Ensures the system is visually appealing, easy to navigate, and functions as expected.

BACKGROUND OF THE SYSTEM

The Elite Tuition Centre's system has been chosen to be created and developed by our group for the CSC435 (Object-Oriented Programming) mini project. This system is a comprehensive solution designed to streamline and simplify the process of enrolling students in various tuition packages. The system allows users to input detailed student information, such as their full name, guardian's contact details, and other essential data required for enrollment. After entering the student's details, users can browse through a variety of available subjects and select the ones most relevant to the student's needs. Along with subject selection, the system also enables users to choose the most appropriate teacher for each subject, ensuring that the student receives high-quality education from qualified instructors.

The system offers a range of predefined tuition packages, each with a fixed price that reflects the cost of the selected subjects and the teachers involved. Users can easily select the package that best fits the student's academic requirements and budget. Once the package and subjects are chosen, the system generates a detailed receipt. This receipt contains all necessary information, including the student's username, date subscription, the teacher assignments, the selected package, the price of the package, the total cost based on the chosen subjects, and the total number of subjects enrolled.

The system ensures an organized and efficient process for managing student enrollments and payments. By generating clear and accurate receipts, it provides a transparent overview of the tuition services provided to each student. This not only helps in keeping track of payments but also assists in managing the teaching staff, ensuring that the students are able to select desired packages and teachers. With its student-friendly interface and streamlined operations, the system enhances the overall experience for both the students and staff of the Elite Tuition Centre, ensuring smooth and effective educational management.

OBJECTIVES OF THE DEVELOPMENT

1. User-Friendly Interface

This system develop a Graphical User Interface (GUI) that simplifies tuition registration for students and ensures the interface is intuitive, visually appealing, and accessible for all students that need helps in finding best tuition that fit with their needs and budgets.

2. Efficient Subject and Teacher Selection

To allow students or guardians to easily browse through available subjects and select the ones most relevant to the student's academic needs, while ensuring that teachers are appropriately matched with subjects.

3. Predefined Tuition Packages and Budget Management

To provide a range of predefined tuition packages, each with a fixed price based on selected subjects and teachers, helping users make informed decisions based on academic needs and budget.

4. Enhanced Teacher Management

To efficiently manage teacher assignments by allowing students or guardians to select teachers based on the subjects and classes they require, ensuring an optimal educational experience.

5. Transparent and Accurate Billing System

To efficiently manage teacher assignments by allowing students or guardians to select teachers based on the subjects and classes they require, ensuring an optimal educational experience.

PLATFORM USED FOR IMPLEMENTATION

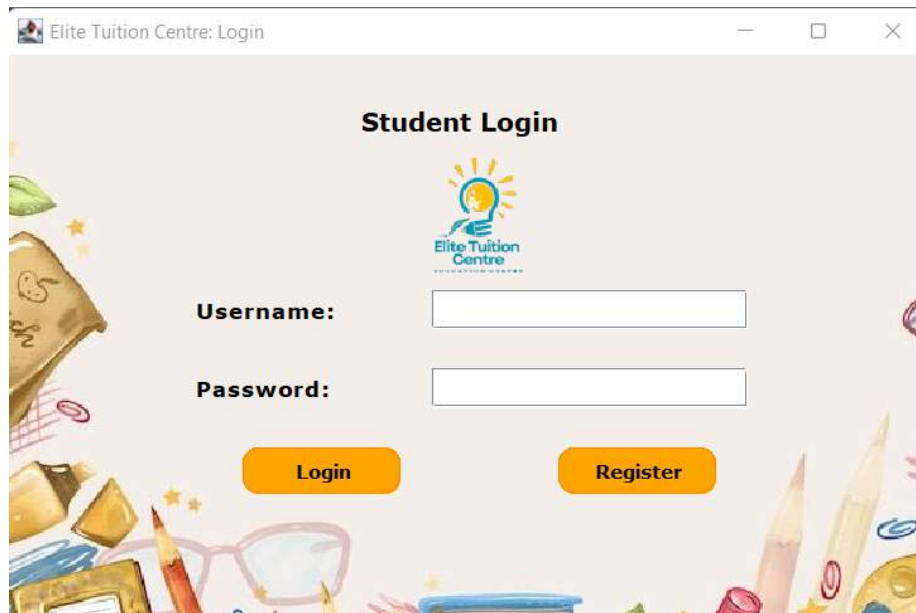
The Elite Tuition Centre: Tuition Registration System is developed using BlueJ, an integrated development environment (IDE) tailored for object-oriented programming. BlueJ is chosen for its simplicity and ease of use, making it an ideal platform for creating a GUI-based application. The system utilizes BlueJ's visual interface to efficiently manage student data, teacher assignments, subject selections, and tuition packages. This graphical user interface (GUI) enhances the user experience, allowing non-technical users to interact with the system seamlessly.

The project is implemented using Java, which enables the development of robust backend logic to handle operations such as user inputs, data processing, and file management. Java's object-oriented principles align well with the system's needs, organizing the core entities like "Main System", "Core System", "Course Subscription", "Information Page", "Rounded Button" and "Print Receipt" into separate classes. This modular approach ensures that the system is easy to maintain and extend. Data is stored in text files, allowing for a lightweight storage solution that eliminates the need for complex databases.

In addition to Java programming, file handling is incorporated for managing the storage and retrieval of student details, teacher assignments, and package information. The system reads and writes data to text files, enabling it to persist information even after the application is closed. The GUI, combined with the object-oriented design and file handling, ensures the Elite Tuition Centre Tuition Registration System operates efficiently, providing a user-friendly and intuitive platform for students.

SCREEN DESIGN AND GUI COMPONENTS USED

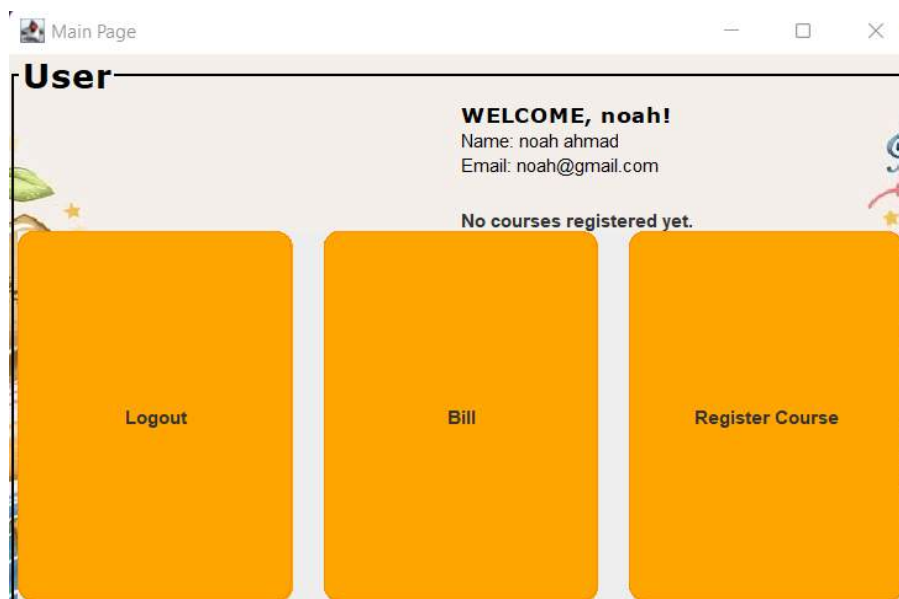
Login Page



The screenshot shows a web browser window titled "Elite Tuition Centre: Login". The page has a light beige background with a decorative border of school supplies like pencils, erasers, and a ruler. In the center, there is a logo for "Elite Tuition Centre" featuring a lightbulb. Below the logo, the text "Student Login" is displayed. There are two input fields: "Username:" and "Password:". Below these fields are two orange buttons: "Login" and "Register".

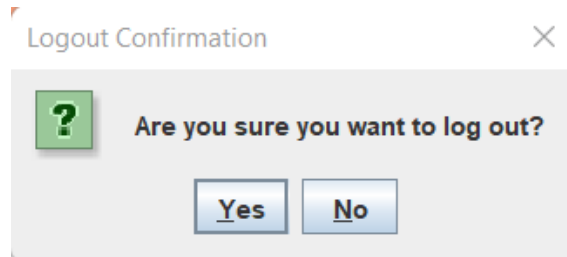
For first-time users, they can click on the "Register" button to be directed to the Student Registration Form. For returning users who have already registered, they can simply enter their username and password, which they provided during the initial registration, and click the "Login" button.

Main Page

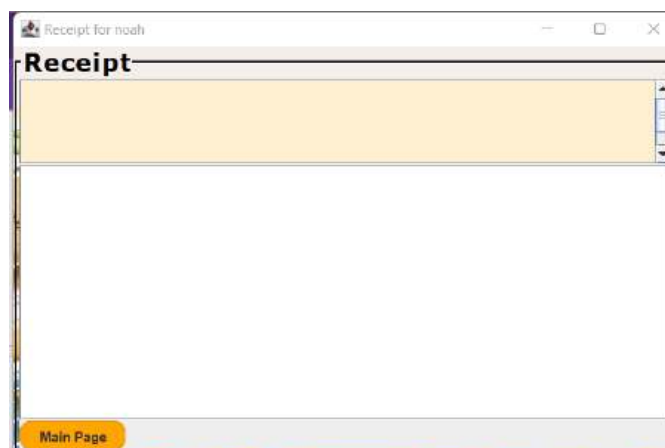


The screenshot shows a web browser window titled "Main Page". The page has a light beige background with a decorative border of school supplies. At the top left, the word "User" is displayed. In the top right, there is a welcome message: "WELCOME, noah!" followed by "Name: noah ahmad" and "Email: noah@gmail.com". Below this, it says "No courses registered yet.". At the bottom, there are three large orange buttons: "Logout", "Bill", and "Register Course".

After the user clicks the "Login" button, they will be directed to the Main Page. This page will display the user's name and email as entered during registration. If it's the user's first time logging in, a message will pop up saying, "No courses registered yet." Below the user's details, there will be three buttons: "Logout," "Bill," and "Register Course." For the "Logout" button, when the user clicks it, a confirmation message will pop up. If the user confirms the logout by clicking "OK", they will be redirected to the login page.

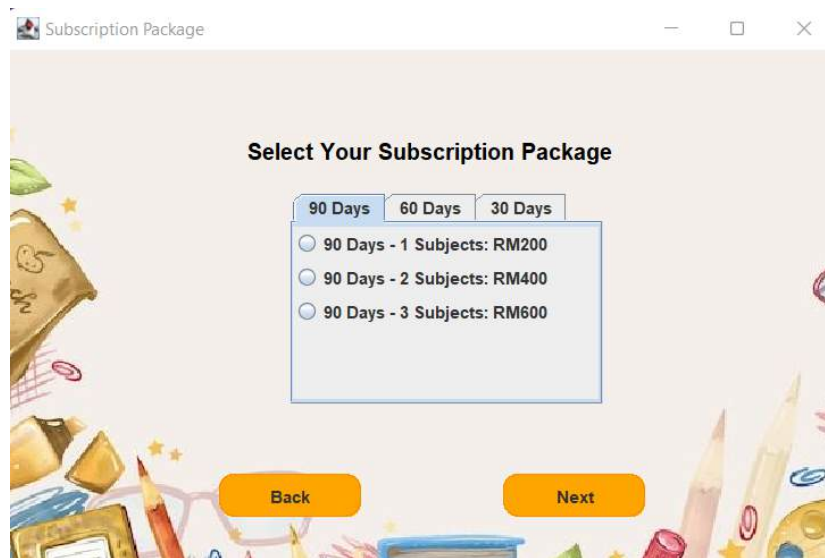


Receipt for user's first time logging



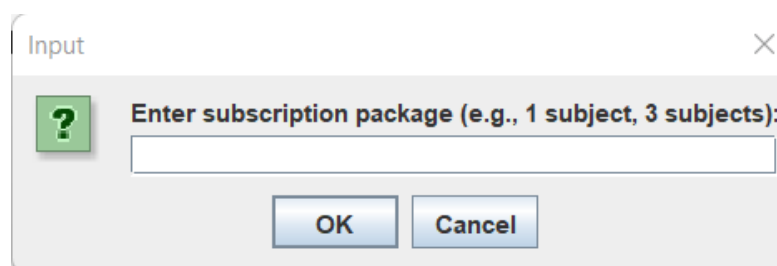
If it's the user's first time logging in and they click the "Bill" button, the system will display a message indicating that no course subscription has been made yet. The "Main Page" button below will redirect the user back to the main page frame.

Subscription Package Frame

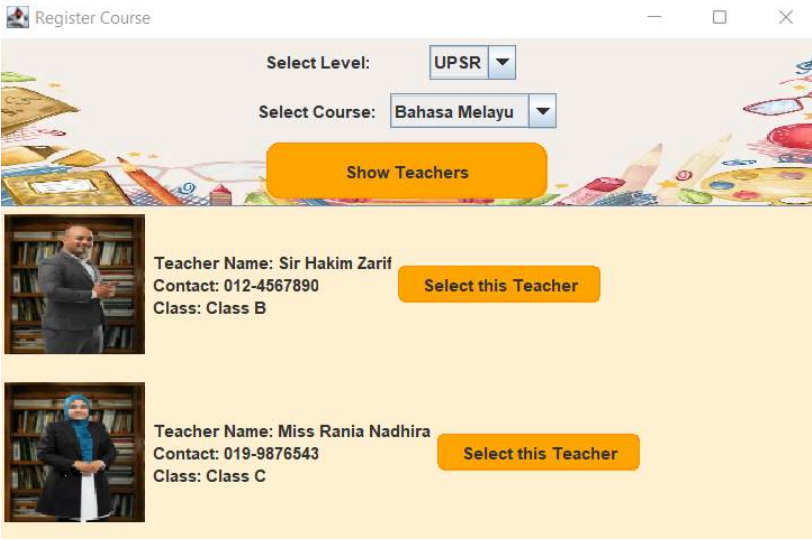
A screenshot of a web application window titled "Subscription Package". The window has a light beige background with a decorative border of school supplies like pencils, erasers, and stars. The main heading is "Select Your Subscription Package". Below it, there are three tabs: "90 Days", "60 Days", and "30 Days". The "90 Days" tab is selected. Under the "90 Days" tab, there are three radio button options: "90 Days - 1 Subjects: RM200", "90 Days - 2 Subjects: RM400", and "90 Days - 3 Subjects: RM600". At the bottom of the window, there are two orange buttons: "Back" and "Next".

When the user clicks the "Register" button, they will be directed to the Subscription Package Frame. In this frame, the user is required to select a package based on the number of subscription days, the number of subjects they wish to take, and their budget that meets their needs. Below, there are two buttons: the "Back" button and the "Next" button. If the user clicks the "Back" button, they will be redirected to the main page, and if they click the "Next" button, they will be directed to the Input Frame, where they will be prompted to enter details for the subjects selected in the package (**only enter the value of subject same like in the package chosen before**)

Input frame

A screenshot of a small dialog box titled "Input". It has a light gray background and a close button (X) in the top right corner. On the left side, there is a green square icon with a white question mark. To the right of the icon, the text "Enter subscription package (e.g., 1 subject, 3 subjects):" is displayed. Below this text is a text input field. At the bottom of the dialog box, there are two buttons: "OK" and "Cancel".

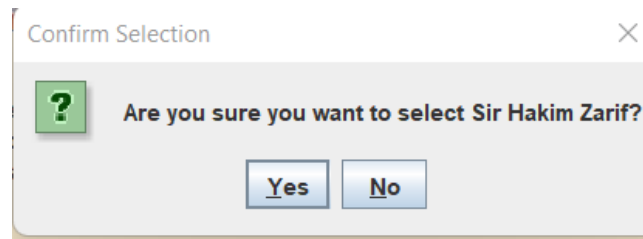
Register course frame



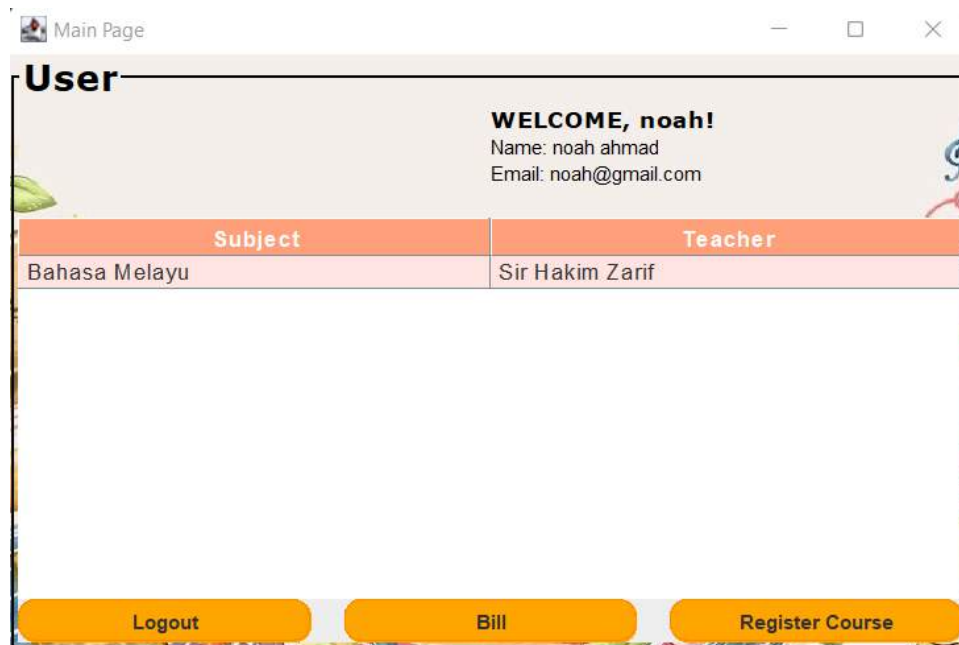
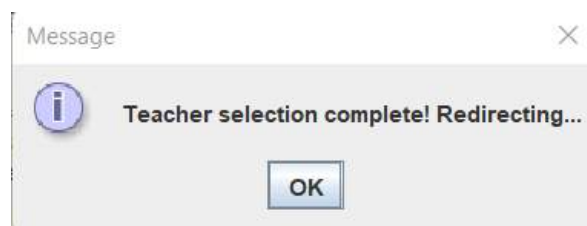
The image shows a software window titled "Register Course". At the top, there are two dropdown menus: "Select Level:" with "UPSR" selected, and "Select Course:" with "Bahasa Melayu" selected. Below these is a large orange button labeled "Show Teachers". The main area of the window displays two teacher profiles. Each profile includes a small photo of the teacher, their name, contact number, and class. To the right of each profile is an orange button labeled "Select this Teacher".

Teacher Name	Contact	Class
Sir Hakim Zarif	012-4567890	Class B
Miss Rania Nadhira	019-9876543	Class C

In this page, the user needs to select the level of their subject (UPSR, PT3, or SPM) and the course (such as Bahasa Melayu, Science, Biology, etc.). After selecting the subject and level, the user must click the "Show Teachers" button to view available teachers and their specific classes. This will allow the user to choose the teacher and class that best suits their preferences.

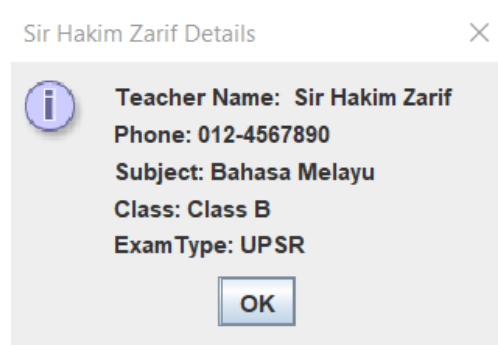


When the user clicks the "Select this Teacher" button, a confirmation message will appear. If the user confirms by clicking "Yes", a message will be displayed below confirming their teacher selection.

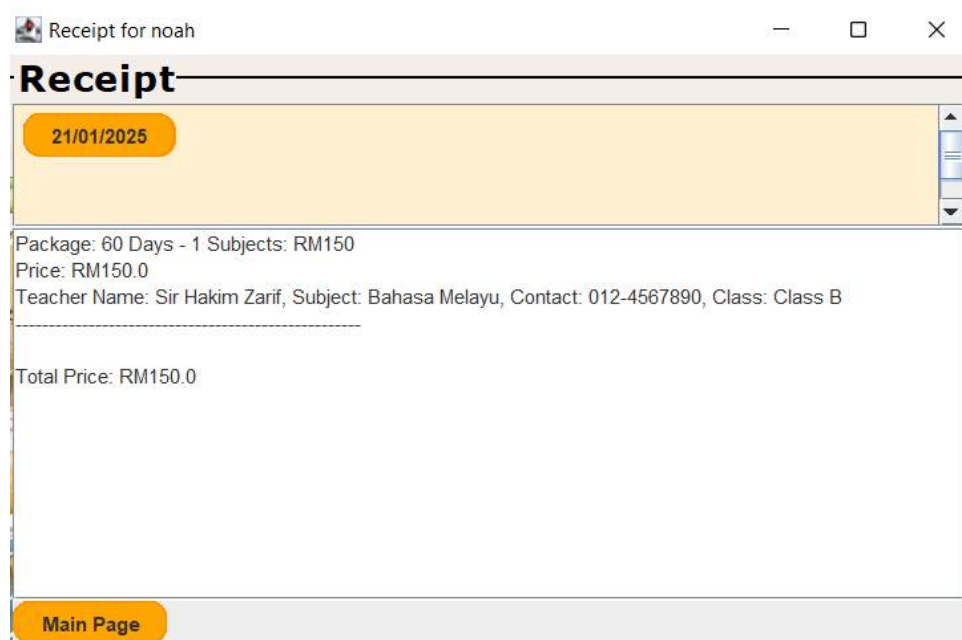


After clicking "OK", the user will be directed to the main page frame. At the top of the frame, the user's name and email address will be displayed. Below that, a table will show the chosen subjects and their corresponding teachers. The user can click on a teacher's name to view specific details about the teacher, including their name, phone number, the subjects they

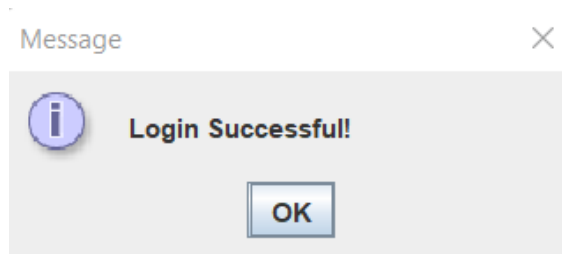
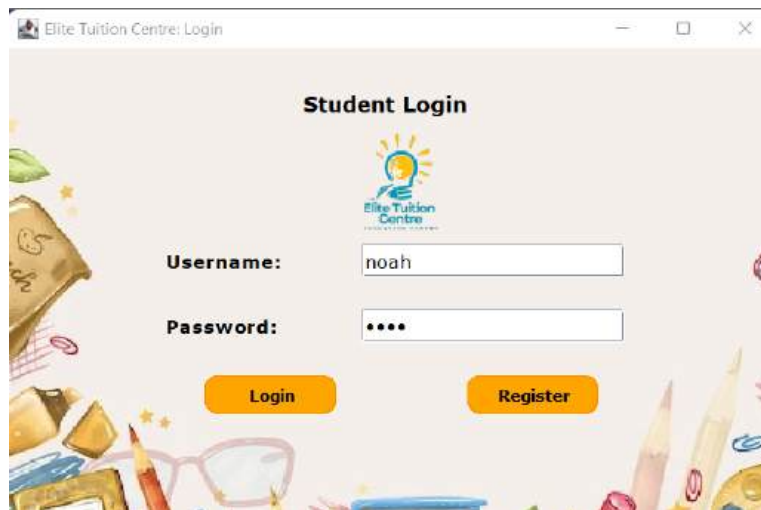
teach, and the exam type for the subject. Below the table, there will be three buttons: "Logout," "Bill," and "Register Course."



Receipt frame



If the user has already confirmed their subscription, they can click on the "Bill" button to view their receipt. The receipt will display a button corresponding to the registration date, which the user can click to view their subscription details. Upon clicking, the package chosen, its price, and a list of teacher details (including teacher name, subject, contact number, and class) for the selected subjects will be shown. The total price of the package will also be displayed, and if the user registers for additional subscription packages, the total price and receipt will be updated accordingly. After reviewing the receipt, the user can click the "Main Page" button to return to the login frame.



When a registered user logs back in, a message will appear confirming their successful login, and they will be directed to the main page.

```
sofea,sofea,sofea@gmail.com,umairah,0173435435,sofea,No courses registered
ilham,ilham,ilham@gmail.com,ahmad,0124325345,ilham,No courses registered, Registered Course: Science, Teacher: Teacher Izzat, Registered
Course: English, Teacher: Teacher Sarah, Registered Course: Science, Teacher: Teacher Rina
siti,siti,siti@gmail.com,hawa,0123435465,siti,No courses registered, Registered Course: Science, Teacher: Teacher Rina, Registered Course:
English, Teacher: Teacher Jack, Registered Course: Science, Teacher: Teacher Faris, Registered Course: Science, Teacher: Teacher Izzat,
Registered Course: Mathematics, Teacher: Madam Marlya
yin,Nurul Ain Safiyyah,icynite@gmail.com,norli,0189778106,yin,No courses registered, Registered Course: Mathematics, Teacher: Miss Raudah,
Registered Course: Bahasa Melayu, Teacher: Miss Rania Nadhira, Registered Course: Chemistry, Teacher: Miss Nurin Alya
noah,noah ahmad,noah@gmail.com,ahmad,0124324345,noah,No courses registered, Registered Course: Bahasa Melayu, Teacher: Sir Hakim Zarif
```

Userdata3.txt

```
Username: yin
Date: 20/01/2025
Package: 30 Days - 1 Subjects: RM100
Price: RM100.0
Teacher Name: Miss Raudah, Subject: Mathematics, Contact: 017-3344556, Class: Class B
-----
Username: yin
Date: 20/01/2025
Package: 30 Days - 1 Subjects: RM100
Price: RM100.0
Teacher Name: Miss Rania Nadhira, Subject: Bahasa Melayu, Contact: 019-9876543, Class: Class C
-----
Username: yin
Date: 21/01/2025
Package: 30 Days - 1 Subjects: RM100
Price: RM100.0
Teacher Name: Miss Nurin Alya, Subject: Chemistry, Contact: 011-3344556, Class: Class B
-----
Username: noah
Date: 21/01/2025
Package: 60 Days - 1 Subjects: RM150
Price: RM150.0
Teacher Name: Sir Hakim Zarif, Subject: Bahasa Melayu, Contact: 012-4567890, Class: Class B
-----
```

Subscription3.txt

```
userdata3.txt  subscription3.txt  Teachers.txt
File Edit View
Miss Ameera Liyana,019-5566778,Science,Class B,C:\\Users\\NURUL NAJWA\\Downloads\\teacher\\TEACHER\\puan10.jpg,PT3
Miss Zahira Elliana,012-1234567,Science,Class C,C:\\Users\\NURUL NAJWA\\Downloads\\teacher\\TEACHER\\puan11.jpg,PT3
Sir Ikhwan Danish,017-4455667,Geography,Class A,C:\\Users\\NURUL NAJWA\\Downloads\\teacher\\TEACHER\\laki14.jpg,PT3
Sir Rafique Azim,018-5544332,Geography,Class B,C:\\Users\\NURUL NAJWA\\Downloads\\teacher\\TEACHER\\laki15.jpg,PT3
Miss Nurin Alya,011-3344556,Geography,Class C,C:\\Users\\NURUL NAJWA\\Downloads\\teacher\\TEACHER\\puan12.jpg,PT3
Sir Badrul Afiq,013-4567890,Bahasa Inggeris,Class A,C:\\Users\\NURUL NAJWA\\Downloads\\teacher\\TEACHER\\laki9.jpg,SPM
Sir Luqman Hakimi,010-9988776,Bahasa Inggeris,Class B,C:\\Users\\NURUL NAJWA\\Downloads\\teacher\\TEACHER\\laki13.jpg,SPM
Madam Zulayka Yasmin,012-5678910,Bahasa Inggeris,Class C,C:\\Users\\NURUL NAJWA\\Downloads\\teacher\\TEACHER\\puan7.jpg,SPM
Madam Hanna Sofea,016-9876543,Bahasa Inggeris,Class D,C:\\Users\\NURUL NAJWA\\Downloads\\teacher\\TEACHER\\puan8.jpg,SPM
Sir Nazril Syahmi,013-3456789,Bahasa Melayu,Class A,C:\\Users\\NURUL NAJWA\\Downloads\\teacher\\TEACHER\\laki11.jpg,SPM
Miss Putri Natasya,017-2345678,Bahasa Melayu,Class B,C:\\Users\\NURUL NAJWA\\Downloads\\teacher\\TEACHER\\puan6.jpg,SPM
Sir Irfan Qayyum,014-6789012,Additional Mathematics,Class A,C:\\Users\\NURUL NAJWA\\Downloads\\teacher\\TEACHER\\laki10.jpg,SPM
Sir Syabil Rayyan,011-2233445,Additional Mathematics,Class B,C:\\Users\\NURUL NAJWA\\Downloads\\teacher\\TEACHER\\laki12.jpg,SPM
Miss Adelia Farhana,018-1122334,Additional Mathematics,Class C,C:\\Users\\NURUL NAJWA\\Downloads\\teacher\\TEACHER\\puan9.jpg,SPM
Sir Ikhwan Danish,017-4455667,Mathematics,Class A,C:\\Users\\NURUL NAJWA\\Downloads\\teacher\\TEACHER\\laki14.jpg,SPM
Sir Luqman Hakimi,010-9988776,Mathematics,Class B,C:\\Users\\NURUL NAJWA\\Downloads\\teacher\\TEACHER\\laki13.jpg,SPM
Sir Rafique Azim,018-5544332,Mathematics,Class C,C:\\Users\\NURUL NAJWA\\Downloads\\teacher\\TEACHER\\laki15.jpg,SPM
Miss Nurin Alya,011-3344556,Mathematics,Class D,C:\\Users\\NURUL NAJWA\\Downloads\\teacher\\TEACHER\\puan12.jpg,SPM
Miss Ameera Liyana,019-5566778,Mathematics,Class E,C:\\Users\\NURUL NAJWA\\Downloads\\teacher\\TEACHER\\puan10.jpg,SPM
Sir Syabil Rayyan,011-2233445,Biology,Class A,C:\\Users\\NURUL NAJWA\\Downloads\\teacher\\TEACHER\\laki12.jpg,SPM
Sir Irfan Qayyum,014-6789012,Biology,Class B,C:\\Users\\NURUL NAJWA\\Downloads\\teacher\\TEACHER\\laki10.jpg,SPM
Miss Zahira Elliana,012-1234567,Biology,Class C,C:\\Users\\NURUL NAJWA\\Downloads\\teacher\\TEACHER\\puan11.jpg,SPM
Miss Adelia Farhana,018-1122334,Physics,Class B,C:\\Users\\NURUL NAJWA\\Downloads\\teacher\\TEACHER\\puan9.jpg,SPM
Madam Hanna Sofea,016-9876543,Physics,Class C,C:\\Users\\NURUL NAJWA\\Downloads\\teacher\\TEACHER\\puan8.jpg,SPM
Sir Ikhwan Danish,017-4455667,Chemistry,Class A,C:\\Users\\NURUL NAJWA\\Downloads\\teacher\\TEACHER\\laki14.jpg,SPM
Miss Nurin Alya,011-3344556,Chemistry,Class B,C:\\Users\\NURUL NAJWA\\Downloads\\teacher\\TEACHER\\puan12.jpg,SPM
```

Teachers.txt

CODING IMPLEMENTATION

1) MainSystem.java

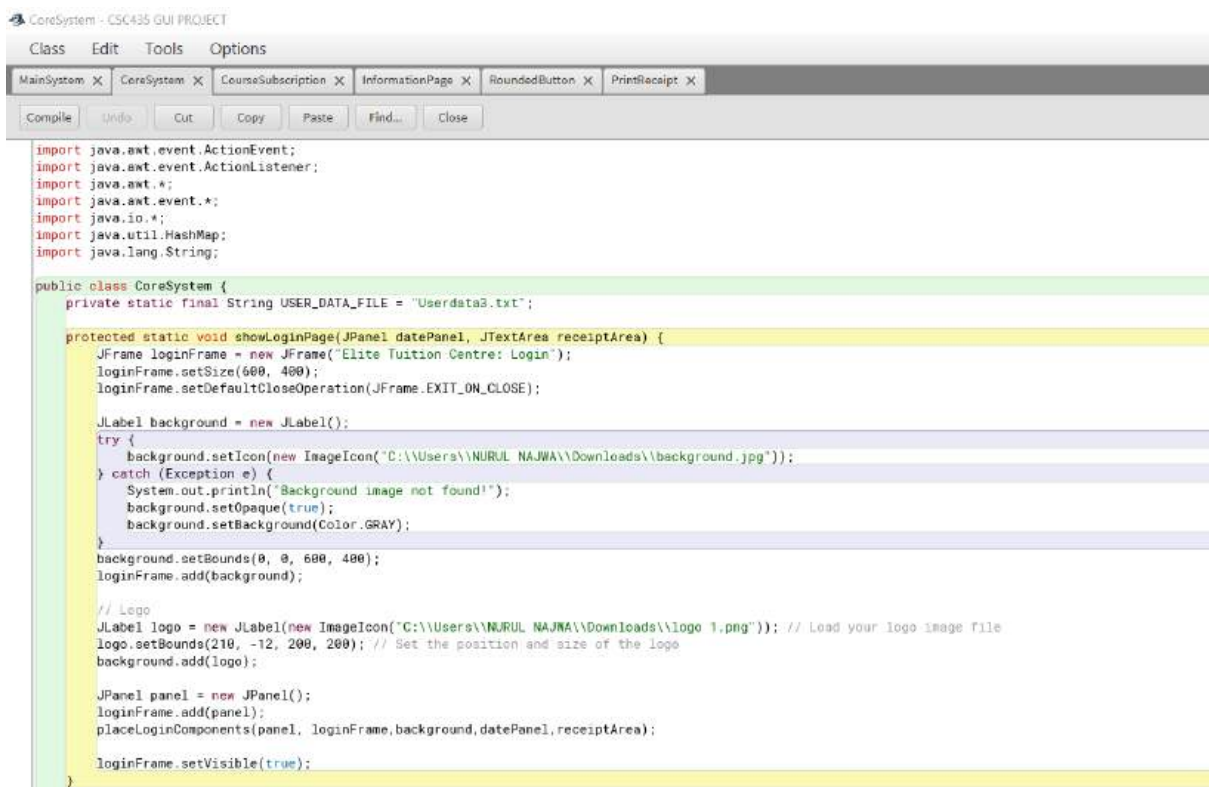


```
import javax.swing.*;
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;
import java.awt.*;
import java.awt.event.*;
import java.io.*;
import java.util.HashMap;
import java.lang.String;

public class MainSystem{
    public static void main(String[] args) {
        JPanel datePanel = new JPanel();
        JTextArea receiptArea = new JTextArea();

        CoreSystem.showLoginPage(datePanel, receiptArea);
    }
}
```

2) CoreSystem.java



```
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;
import java.awt.*;
import java.awt.event.*;
import java.io.*;
import java.util.HashMap;
import java.lang.String;

public class CoreSystem {
    private static final String USER_DATA_FILE = "Userdata3.txt";

    protected static void showLoginPage(JPanel datePanel, JTextArea receiptArea) {
        JFrame loginFrame = new JFrame("Elite Tuition Centre: Login");
        loginFrame.setSize(600, 400);
        loginFrame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);

        JLabel background = new JLabel();
        try {
            background.setIcon(new ImageIcon("C:\\Users\\NURUL NAJWA\\Downloads\\background.jpg"));
        } catch (Exception e) {
            System.out.println("Background image not found!");
            background.setOpaque(true);
            background.setBackground(Color.GRAY);
        }
        background.setBounds(0, 0, 600, 400);
        loginFrame.add(background);

        // Logo
        JLabel logo = new JLabel(new ImageIcon("C:\\Users\\NURUL NAJWA\\Downloads\\logo 1.png")); // Load your logo image file
        logo.setBounds(210, -12, 200, 200); // Set the position and size of the logo
        background.add(logo);

        JPanel panel = new JPanel();
        loginFrame.add(panel);
        placeLoginComponents(panel, loginFrame, background, datePanel, receiptArea);

        loginFrame.setVisible(true);
    }
}
```



```
private static void placeLoginComponents(JPanel panel, JFrame loginFrame, JLabel background, JPanel datePanel, JTextArea receiptArea) {
    panel.setLayout(null);

    JLabel titleLabel = new JLabel("Student Login");
    titleLabel.setBounds(225, 30, 200, 25);
    titleLabel.setForeground(Color.BLACK);
    titleLabel.setFont(new Font("Verdana", Font.BOLD, 16));
    background.add(titleLabel);

    JLabel userLabel = new JLabel("Username:");
    userLabel.setBounds(120, 150, 100, 25);
    userLabel.setForeground(Color.BLACK);
    userLabel.setFont(new Font("Verdana", Font.BOLD, 14));
    panel.add(userLabel);
    background.add(userLabel);

    JTextField userText = new JTextField(20);
    userText.setBounds(270, 150, 200, 25);
    userText.setForeground(Color.BLACK);
    userText.setFont(new Font("Verdana", Font.PLAIN, 14));
    panel.add(userText);
    background.add(userText);

    JLabel passwordLabel = new JLabel("Password:");
    passwordLabel.setBounds(120, 200, 100, 25);
    panel.add(passwordLabel);
    passwordLabel.setForeground(Color.BLACK);
    passwordLabel.setFont(new Font("Verdana", Font.BOLD, 14));
    background.add(passwordLabel);

    JPasswordField passwordText = new JPasswordField(20);
    passwordText.setBounds(270, 200, 200, 25);
    panel.add(passwordText);
    passwordText.setForeground(Color.BLACK);
    passwordText.setFont(new Font("Verdana", Font.PLAIN, 14));
    background.add(passwordText);

    RoundedButton loginButton = new RoundedButton("Login", 20);
    loginButton.setBounds(150, 250, 100, 30);
    panel.add(loginButton);

```

```
loginButton.setForeground(Color.BLACK);
loginButton.setFont(new Font("Verdana", Font.BOLD, 12));
background.add(loginButton);

RoundedButton registerButton = new RoundedButton("Register", 20);
registerButton.setBounds(350, 250, 100, 30);
panel.add(registerButton);
registerButton.setForeground(Color.BLACK);
registerButton.setFont(new Font("Verdana", Font.BOLD, 12));
background.add(registerButton);

```

```
loginButton.addActionListener(new ActionListener() {
    @Override
    public void actionPerformed(ActionEvent e) {
        String username = userText.getText();
        String password = new String(passwordText.getPassword());
        if (validateLogin(username, password)) {
            JOptionPane.showMessageDialog(null, "Login Successful!");
            InformationPage.showMainPage(username, datePanel, receiptArea);
            loginFrame.dispose();
        } else {
            JOptionPane.showMessageDialog(null, "Invalid Username or Password.");
        }
    }
});

```

```
registerButton.addActionListener(new ActionListener() {
    @Override
    public void actionPerformed(ActionEvent e) {
        String username = userText.getText();
        String password = new String(passwordText.getPassword());
        if (registerUser(username, password)) {
            Register(datePanel, receiptArea);
            loginFrame.dispose();
        } else {
            JOptionPane.showMessageDialog(null, "Registration Failed. Username may already exist.");
        }
    }
});

```

```

    });
}

public static boolean validateLogin(String username, String password) {
    try (BufferedReader reader = new BufferedReader(new FileReader(USER_DATA_FILE))) {
        String line;
        while ((line = reader.readLine()) != null) {
            String[] parts = line.split(",");
            if (parts.length >= 6 && parts[0].equals(username) && parts[5].equals(password)) {
                return true; // Username and password match
            }
        }
    } catch (IOException e) {
        e.printStackTrace();
    }
    return false; // No match found
}

private static boolean registerUser(String username, String password) {
    try (BufferedReader reader = new BufferedReader(new FileReader(USER_DATA_FILE))) {
        String line;
        while ((line = reader.readLine()) != null) {
            String[] parts = line.split(",");
            if (parts[0].equals(username)) {
                return false; // Username already exists
            }
        }
    } catch (IOException e) {
        // File may not exist yet, so proceed
    }
    return true;
}

private static void Register(JPanel datePanel, JTextArea receiptArea){
    JFrame frame = new JFrame("Student Registration");
    frame.setSize(615, 430);
    frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
    frame.setLayout(null);
}

```

```

JLabel background = new JLabel();
try {
    background.setIcon(new ImageIcon("C:\\Users\\NURUL NAJWA\\Downloads\\background.jpg"));
} catch (Exception e) {
    System.out.println("Background image not found!");
    background.setOpaque(true);
    background.setBackground(Color.GRAY);
}
background.setBounds(0, 0, 600, 400); // Set background size

// Now add the background panel to the frame
frame.add(background);

// Title Label
JLabel titleLabel = new JLabel("Student Registration");
titleLabel.setBounds(200, 20, 200, 30);
titleLabel.setHorizontalAlignment(SwingConstants.CENTER);
titleLabel.setForeground(Color.BLACK);
titleLabel.setFont(new Font("Verdana", Font.BOLD, 16));
background.add(titleLabel);

// Full Name
JLabel nameLabel = new JLabel("Full Name:");
nameLabel.setBounds(120, 80, 100, 25);
nameLabel.setForeground(Color.BLACK);
nameLabel.setFont(new Font("Verdana", Font.BOLD, 12));
background.add(nameLabel);
JTextField nameField = new JTextField();
nameField.setBounds(270, 80, 200, 25);
nameField.setForeground(Color.BLACK);
nameField.setFont(new Font("Verdana", Font.PLAIN, 12));
background.add(nameField);

// Email
JLabel emailLabel = new JLabel("Email:");
emailLabel.setBounds(120, 120, 100, 25);
emailLabel.setForeground(Color.BLACK);
emailLabel.setFont(new Font("Verdana", Font.BOLD, 12));
background.add(emailLabel);

```

```

JTextField emailField = new JTextField();
emailField.setBounds(270, 120, 200, 25);
emailField.setForeground(Color.BLACK);
emailField.setFont(new Font("Verdana", Font.PLAIN, 12));
background.add(emailField);

// Username
JLabel userLabel = new JLabel("Username:");
userLabel.setBounds(120, 160, 100, 25);
userLabel.setForeground(Color.BLACK);
userLabel.setFont(new Font("Verdana", Font.BOLD, 12));
background.add(userLabel);

JTextField usernameText = new JTextField();
usernameText.setBounds(270, 160, 200, 25);
usernameText.setForeground(Color.BLACK);
usernameText.setFont(new Font("Verdana", Font.PLAIN, 12));
background.add(usernameText);

// Password
JLabel passwordLabel = new JLabel("Password:");
passwordLabel.setBounds(120, 200, 100, 25);
passwordLabel.setForeground(Color.BLACK);
passwordLabel.setFont(new Font("Verdana", Font.BOLD, 12));
background.add(passwordLabel);

JPasswordField passwordField = new JPasswordField();
passwordField.setBounds(270, 200, 200, 25);
passwordField.setForeground(Color.BLACK);
passwordField.setFont(new Font("Verdana", Font.PLAIN, 12));
background.add(passwordField);

// Confirm Password
JLabel confirmPasswordLabel = new JLabel("Confirm Password:");
confirmPasswordLabel.setBounds(120, 240, 150, 25);
confirmPasswordLabel.setForeground(Color.BLACK);
confirmPasswordLabel.setFont(new Font("Verdana", Font.BOLD, 12));
background.add(confirmPasswordLabel);

JPasswordField confirmPasswordField = new JPasswordField();
confirmPasswordField.setBounds(270, 240, 200, 25);
confirmPasswordField.setForeground(Color.BLACK);
confirmPasswordField.setFont(new Font("Verdana", Font.PLAIN, 12));
background.add(confirmPasswordField);

// Guardian Name
JLabel guardianLabel = new JLabel("Guardian's Name:");
guardianLabel.setBounds(120, 280, 150, 25);
guardianLabel.setForeground(Color.BLACK);
guardianLabel.setFont(new Font("Verdana", Font.BOLD, 12));
background.add(guardianLabel);

JTextField guardianField = new JTextField();
guardianField.setBounds(270, 280, 200, 25);
guardianField.setForeground(Color.BLACK);
guardianField.setFont(new Font("Verdana", Font.PLAIN, 12));
background.add(guardianField);

// Guardian Phone Number
JLabel guardianPhoneLabel = new JLabel("Guardian's Phone:");
guardianPhoneLabel.setBounds(120, 320, 150, 25);
guardianPhoneLabel.setForeground(Color.BLACK);
guardianPhoneLabel.setFont(new Font("Verdana", Font.BOLD, 12));
background.add(guardianPhoneLabel);

JTextField guardianPhoneField = new JTextField();
guardianPhoneField.setBounds(270, 320, 200, 25);
guardianPhoneField.setForeground(Color.BLACK);
guardianPhoneField.setFont(new Font("Verdana", Font.PLAIN, 12));
background.add(guardianPhoneField);

// Back Button
RoundedButton backButton = new RoundedButton("Back", 20);
backButton.setBounds(120, 360, 100, 30);
background.add(backButton);
backButton.setForeground(Color.BLACK);
backButton.setFont(new Font("Verdana", Font.BOLD, 12));

```

```

// Next Button
RoundedButton nextButton = new RoundedButton("Next", 20);
nextButton.setBounds(370, 360, 180, 30);
background.add(nextButton);
nextButton.setForeground(Color.BLACK);
nextButton.setFont(new Font("Verdana", Font.BOLD, 12));

backButton.addActionListener(e -> {
    frame.dispose();
    showLoginPage(datePanel, receiptArea); // Go back to the previous frame
});

nextButton.addActionListener(new ActionListener() {
    @Override
    public void actionPerformed(ActionEvent e) {
        String password = new String(passwordField.getPassword());
        String confirmPassword = new String(confirmPasswordField.getPassword());
        String username = usernameText.getText().trim();
        String name = nameField.getText().trim();
        String email = emailField.getText().trim();
        String guardian = guardianField.getText().trim();
        String guardianPhone = guardianPhoneField.getText().trim();

        if (username.isEmpty() || name.isEmpty() || email.isEmpty() || password.isEmpty() || confirmPassword.isEmpty() || guardian.isEmpty() || guardianPhone.isEmpty()) {
            JOptionPane.showMessageDialog(frame, "Please fill in all fields!", "Error", JOptionPane.ERROR_MESSAGE);
        } else if (!password.equals(confirmPassword)) {
            JOptionPane.showMessageDialog(frame, "Passwords do not match!", "Error", JOptionPane.ERROR_MESSAGE);
        } else {
            try (BufferedWriter writer = new BufferedWriter(new FileWriter(USER_DATA_FILE, true))) {
                writer.write(username + "," + name + "," + email + "," + guardian + "," + guardianPhone + "," + password + ",No courses registered\n");
                writer.flush(); // Ensures data is written to the file
            } catch (IOException f) {
                f.printStackTrace();
            }

            frame.dispose();
            InformationPage.showMainPage(username, datePanel, receiptArea); // Redirect to the main page
        }
    }
});
frame.setVisible(true);
}
}

```

3) CourseSubscription.java

```

import javax.swing.*;
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;
import java.awt.*;
import java.awt.event.*;
import java.io.*;
import java.util.HashMap;
import java.lang.String;
import java.util.ArrayList;
import java.time.LocalDate;
import java.time.format.DateTimeFormatter;
import javax.swing.JTextArea;

public class CourseSubscription{
    private static ArrayList<String[]> selectedTeachers = new ArrayList<>();
    private static int subjectsToSelect;
    private static int currentSelectionCount = 0;
    private static final String USER_DATA_FILE = "userdata3.txt";

    //code for choosing package interface
    protected static void subscription(String username, JFrame mainFrame, JPanel datePanel, JTextArea receiptArea){
        if (mainFrame != null) {
            mainFrame.dispose();
        }

        JFrame frame = new JFrame("Subscription Package");
        frame.setSize(600, 400);
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        frame.setLayout(null);
    }
}

```

```

// Background image
JLabel background = new JLabel();
try {
    background.setIcon(new ImageIcon("C:\\Users\\user\\BlueJ Project\\LabProject\\image100P.jpg"));
} catch (Exception e) {
    System.out.println("Background image not found!");
    background.setOpaque(true);
    background.setBackground(Color.GRAY);
}

background.setBounds(0, 0, 600, 400);
frame.add(background);

// Title
JLabel titleLabel = new JLabel("Select Your Subscription Package");
titleLabel.setBounds(170, 60, 300, 25);
titleLabel.setForeground(Color.BLACK);
titleLabel.setFont(new Font("Arial", Font.BOLD, 16));
background.add(titleLabel);

// Tabbed Pane for Packages
JTabbedPane tabbedPane = new JTabbedPane();
tabbedPane.setBounds(200, 100, 220, 150);
background.add(tabbedPane);

// Panel for 90 Days Packages
JPanel panel90Days = createPackagePanel(new String[]{
    "90 Days - 1 Subjects: RM200",
    "90 Days - 2 Subjects: RM400",
    "90 Days - 3 Subjects: RM600"
});

// Panel for 60 Days Packages
JPanel panel60Days = createPackagePanel(new String[]{
    "60 Days - 1 Subjects: RM150",
    "60 Days - 2 Subjects: RM300",
    "60 Days - 3 Subjects: RM450"
});

// Panel for 30 Days Packages
JPanel panel30Days = createPackagePanel(new String[]{
    "30 Days - 1 Subjects: RM100",
    "30 Days - 2 Subjects: RM200",
    "30 Days - 3 Subjects: RM300"
});

// Add Panels to Tabbed Pane
tabbedPane.addTab("90 Days", panel90Days);
tabbedPane.addTab("60 Days", panel60Days);
tabbedPane.addTab("30 Days", panel30Days);

// Buttons
RoundedButton backButton = new RoundedButton("Back", 20);
backButton.setBounds(150, 300, 100, 30);
background.add(backButton);

RoundedButton nextButton = new RoundedButton("Next", 20);
nextButton.setBounds(350, 300, 100, 30);
background.add(nextButton);

```

```

// Action for Back Button
backButton.addActionListener(e -> {
    frame.dispose();
    InformationPage.showMainPage(username, datePanel, receiptArea); // Go back to the previous frame
});

// Action for Next Button
nextButton.addActionListener(e -> {
    // Check if the user already has a selected package
    if (isSubscriptionAlreadySelected(username)) {
        int option = JOptionPane.showConfirmDialog(
            frame,
            "You have already selected a subscription package. Do you want to subscribe to a different one?",
            "Confirm Subscription Change",
            JOptionPane.YES_NO_OPTION,
            JOptionPane.QUESTION_MESSAGE
        );

        // If the user selects 'No', do nothing (stay on the current screen)
        if (option == JOptionPane.NO_OPTION) {
            return;
        }

        // Get selected package
        String selectedPackage = getSelectedPackage(panel90Days);

        if (selectedPackage == null) {
            selectedPackage = getSelectedPackage(panel60Days);
        }

        if (selectedPackage == null) {
            selectedPackage = getSelectedPackage(panel30Days);
        }

        if (selectedPackage == null) {
            JOptionPane.showMessageDialog(frame, "Please select a subscription package!", "Error", JOptionPane.ERROR_MESSAGE);
        } else {
            // Define the price based on the selected package
            double price = getPriceForPackage(selectedPackage);

            if (price == -1) {
                JOptionPane.showMessageDialog(frame, "Invalid package selection!", "Error", JOptionPane.ERROR_MESSAGE);
                return;
            }

            String date = getDaySubs(selectedPackage);
            JOptionPane.showMessageDialog(frame, "Package selection saved!");
            frame.dispose();
            confirmation(username, selectedPackage, price, mainFrame, date, datePanel, receiptArea); // Proceed to the next frame
        }
    }
});

frame.setVisible(true);
}

private static boolean isSubscriptionAlreadySelected(String username) {
    try (BufferedReader reader = new BufferedReader(new FileReader("subscription3.txt"))) {
        String line;
        while ((line = reader.readLine()) != null) {
            // Check if the line contains the username (meaning they have already selected a package)
            if (line.contains(username)) {
                return true;
            }
        }

        catch (IOException e) {
            e.printStackTrace();
        }

        return false;
    }

private static String getDaySubs (String selectedPackage){
    LocalDate date = LocalDate.now();
    DateTimeFormatter formatter = DateTimeFormatter.ofPattern("dd/MM/yyyy");
    String datePackageSelected = date.format(formatter);
    return datePackageSelected;
}

private static void confirmation(String username, String selectedPackage, double price, JFrame mainFrame, String date, JPanel datePanel, JTextArea receiptArea){
    String subscriptionPackage = JOptionPane.showInputDialog(
        "Enter subscription package (e.g., 1 subject, 3 subjects):");

    try {
        subjectsToSelect = Integer.parseInt(subscriptionPackage.split(" ")[0]);
    } catch (Exception e) {
        JOptionPane.showMessageDialog(null, "Invalid input. Defaulting to 1 subject.");
        subjectsToSelect = 1; // Default to 1 subject
    }
}

```



```

        showCourseRegistrationPage(username, selectedPackage, price, mainFrame, date, datePanel, receiptArea);
    }

    //code for choose subject and teacher interface
    private static void showCourseRegistrationPage(String username, String selectedPackage, double price, JFrame mainFrame, String date, JPanel datePanel, JTextArea receiptArea) {
        JFrame courseFrame = new JFrame("Register Course");
        courseFrame.setSize(600, 400);
        courseFrame.setDefaultCloseOperation(JFrame.DISPOSE_ON_CLOSE);
        courseFrame.setLayout(new BorderLayout());

        JPanel selectionPanel = new JPanel(new GridBagLayout()) {
            private Image backgroundImage = new ImageIcon("C:\\Users\\user\\BlueJ Project\\LabProject\\image\\00P.jpg").getImage();

            @Override
            protected void paintComponent(Graphics g) {
                super.paintComponent(g);
                // Draw the background image
                g.drawImage(backgroundImage, 0, 0, getWidth(), getHeight(), this);
            }
        };

        GridBagConstraints gbc = new GridBagConstraints();
        gbc.insets = new Insets(5, 5, 5, 5); // Add padding around components

        JPanel panel = new JPanel();
        courseFrame.add(panel);
        panel.setLayout(null);

        JLabel selectLevelLabel = new JLabel("Select Level:");
        String[] levels = {"UPSR", "FIS", "SPM"};
        JComboBox<String> levelComboBox = new JComboBox<>(levels);

        JLabel courseLabel = new JLabel("Select Course:");
        courseLabel.setBounds(10, 20, 150, 25);
        panel.add(courseLabel);

        String[] courses = {"Bahasa Melayu", "Bahasa Inggeris", "Mathematics", "Science", "Geography", "Addmath", "Biology", "Physics", "Chemistry"};
        JComboBox<String> courseBox = new JComboBox<>(courses);
        courseBox.setBounds(150, 20, 200, 25);
        panel.add(courseBox);

        RoundedButton showTeachersButton = new RoundedButton("Show Teachers", 20);

        // Add components to selectionPanel
        gbc.gridx = 0; gbc.gridy = 0;
        selectionPanel.add(selectLevelLabel, gbc);

        gbc.gridx = 1;
        selectionPanel.add(levelComboBox, gbc);

        gbc.gridx = 0; gbc.gridy = 1;
        selectionPanel.add(courseLabel, gbc);

        gbc.gridx = 1;
        selectionPanel.add(courseBox, gbc);

        gbc.gridx = 0; gbc.gridy = 2; gbc.gridwidth = 2;
        showTeachersButton.setPreferredSize(new Dimension(200, 40));
        selectionPanel.add(showTeachersButton, gbc);

        courseFrame.add(selectionPanel, BorderLayout.NORTH);

        JPanel teacherPanel = new JPanel();
        teacherPanel.setLayout(new BoxLayout(teacherPanel, BoxLayout.Y_AXIS));
        teacherPanel.setBackground(new Color(255, 240, 210));
        JScrollPane teacherScrollPane = new JScrollPane(teacherPanel);
        courseFrame.add(teacherScrollPane, BorderLayout.CENTER);

        ArrayList<String[]> selectedTeachers = new ArrayList<>();
        String[][] teacherDetails = readTeacherDataFromFile("C:\\Users\\user\\BlueJ Project\\First1\\Teachers.txt");

        showTeachersButton.addActionListener(e -> {
            String selectedLevel = (String) levelComboBox.getSelectedItem();
            String selectedSubject = (String) courseBox.getSelectedItem();

            teacherPanel.removeAll();
            boolean teacherFound = false;

            if (selectedTeachers.isEmpty()) {
                currentSelectionCount = 0;
            }

            for (String[] teacher : teacherDetails) {
                if (teacher.length > 5 && teacher[2] != null && teacher[5] != null && teacher[2].equals(selectedSubject) && teacher[5].equals(selectedLevel)) {
                    teacherFound = true;

                    JPanel teacherProfilePanel = new JPanel(new FlowLayout(FlowLayout.LEFT));
                    teacherProfilePanel.setBackground(new Color(255, 240, 210));

                    // Load the teacher's image
                    JLabel imageLabel = new JLabel();
                    try {
                        ImageIcon icon = new ImageIcon(teacher[4]); // teacher[4] is the path to the image
                        Image scaledImage = icon.getImage().getScaledInstance(100, 100, Image.SCALE_SMOOTH); // Resize image
                        imageLabel.setIcon(new ImageIcon(scaledImage));
                    } catch (Exception ex) {
                        System.out.println("Image not found for " + teacher[0]);
                        imageLabel.setText("No Image"); // Fallback if image is missing
                    }

                    JPanel teacherDetailsPanel = new JPanel();
                    teacherDetailsPanel.setBackground(new Color(255, 240, 210));
                    teacherDetailsPanel.setLayout(new BoxLayout(teacherDetailsPanel, BoxLayout.Y_AXIS));
                    teacherDetailsPanel.add(new JLabel("Teacher Name: " + teacher[0]));
                }
            }

            teacherPanel.add(teacherProfilePanel);
            teacherPanel.add(teacherDetailsPanel);
        });
    }

```

```

teacherDetailsPanel.add(new JLabel("Contact: " + teacher[1]));
teacherDetailsPanel.add(new JLabel("Class: " + teacher[3]));

RoundedButton selectButton = new RoundedButton("Select this Teacher",10);
selectButton.addActionListener(evt -> {
    int choice = JOptionPane.showConfirmDialog(courseFrame,
        "Are you sure you want to select " + teacher[0] + "?",
        "Confirm Selection", JOptionPane.YES_NO_OPTION);

    if (choice == JOptionPane.YES_OPTION) {
        selectedTeachers.add(teacher);
        currentSelectionCount++;
        JOptionPane.showMessageDialog(courseFrame, "You selected: " + teacher[0]);

        // Save the current course and teacher to the user data file
        updateUserData(username, (String) courseBox.getSelectedItem(), teacher);

        if (currentSelectionCount < subjectsToSelect) {
            JOptionPane.showMessageDialog(courseFrame,
                "Please select a teacher for the next subject.");
            teacherPanel.removeAll();
            teacherPanel.revalidate();
            teacherPanel.repaint();
        }
    } else {
        JOptionPane.showMessageDialog(courseFrame, "Teacher selection complete! Redirecting...");
        PrintReceipt.saveSubscriptionDetails(username, selectedPackage, price, new String[]{(String) courseBox.getSelectedItem(), selectedTeachers.toArray(new String[0]), date, datePanel, receiptArea);
        courseFrame.dispose();
        mainFrame.dispose();
        InformationPage.showMainPage(username, datePanel, receiptArea); // Direct to MainPage
    }
});

teacherProfilePanel.add(imageLabel);
teacherProfilePanel.add(teacherDetailsPanel);
teacherProfilePanel.add(selectButton);

teacherPanel.add(teacherProfilePanel);
}

if (!teacherFound) {
    teacherPanel.add(new JLabel("No teachers found for the selected level and subject."));
}

teacherPanel.revalidate();
teacherPanel.repaint();
});

courseFrame.setVisible(true);
}

private static String[][] readTeacherDataFromFile(String filePath) {
    String[][] teacherDetails = new String[100][6]; // Assuming max 100 teachers
    int index = 0;

    try (BufferedReader br = new BufferedReader(new FileReader(filePath))) {
        String line;
        while ((line = br.readLine()) != null) {
            String[] teacherData = line.split(","); // Split by commas
            teacherDetails[index] = teacherData;
            index++;
        }
    }

    catch (IOException e) {
        e.printStackTrace();
    }
}

```



```

    }

    // Return the array with the teacher data
    return teacherDetails;
}

// Method to determine price based on selected package
private static double getPriceForPackage(String selectedPackage) {
    if (selectedPackage.contains("90 Days") && selectedPackage.contains("1 Subjects")) return 200;
    if (selectedPackage.contains("90 Days") && selectedPackage.contains("2 Subjects")) return 400;
    if (selectedPackage.contains("90 Days") && selectedPackage.contains("3 Subjects")) return 600;
    if (selectedPackage.contains("60 Days") && selectedPackage.contains("1 Subjects")) return 150;
    if (selectedPackage.contains("60 Days") && selectedPackage.contains("2 Subjects")) return 300;
    if (selectedPackage.contains("60 Days") && selectedPackage.contains("3 Subjects")) return 450;
    if (selectedPackage.contains("30 Days") && selectedPackage.contains("1 Subjects")) return 100;
    if (selectedPackage.contains("30 Days") && selectedPackage.contains("2 Subjects")) return 200;
    if (selectedPackage.contains("30 Days") && selectedPackage.contains("3 Subjects")) return 300;

    return -1; // Invalid selection
}

private static void updateUserData(String username, String course, String[] teacher) {
    File file = new File(USER_DATA_FILE);
    File tempFile = new File("temp_userdata.txt");

    try (BufferedReader reader = new BufferedReader(new FileReader(file));
        BufferedWriter writer = new BufferedWriter(new FileWriter(tempFile))) {
        String line;
        boolean userFound = false;

        while ((line = reader.readLine()) != null) {
            if (line.startsWith(username + ",")) {
                // Append course and teacher information to the user's line
                line += ", Registered Course: " + course + ", Teacher: " + teacher[0];
                userFound = true;
            }
            writer.write(line + "\n");
        }

        if (!userFound) {
            // Handle the case where the user does not exist in the file
            writer.write(username + ", Registered Course: " + course + ", Teacher: " + teacher[0] + "\n");
        }
    } catch (IOException e) {
        e.printStackTrace();
    }

    // Replace the original file with the updated file
    if (file.delete()) {
        tempFile.renameTo(file);
    } else {
        System.out.println("Error replacing the user data file.");
    }
}

//method to create radio button packages
private static JPanel createPackagePanel(String[] packages) {
    JPanel panel = new JPanel();
    panel.setLayout(new BoxLayout(panel, BoxLayout.Y_AXIS));
    ButtonGroup buttonGroup = new ButtonGroup();

    for (String pkg : packages) {
        JRadioButton radioButton = new JRadioButton(pkg);
        radioButton.setAlignmentX(Component.LEFT_ALIGNMENT);
        buttonGroup.add(radioButton);
        panel.add(radioButton);
    }

    return panel;
}

// Method to get the selected package from a panel
private static String getSelectedPackage(JPanel panel) {
    Component[] components = panel.getComponents();
    for (Component component : components) {
        if (component instanceof JRadioButton) {
            JRadioButton radioButton = (JRadioButton) component;
            if (radioButton.isSelected()) {
                return radioButton.getText();
            }
        }
    }
    return null;
}

```

4) PrintReceipt.java

```
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;
import java.util.List;
import java.awt.event.*;
import java.awt.*;
import java.io.*;
import java.util.HashMap;
import java.lang.String;
import java.io.BufferedWriter;
import java.io.FileWriter;
import java.io.IOException;
import java.time.LocalDate;
import java.time.Period;
import java.time.format.DateTimeFormatter;
import java.lang.String;
import java.util.ArrayList;
import javax.swing.JTextArea;
import javax.swing.border.TitledBorder;

public class PrintReceipt {
    //code shows receipt interface
    protected static void generateReceipt(String username, String userData, String[][] coursesAndTeachers, JPanel datePanel, JTextArea receiptArea) {
        JFrame frame = new JFrame("Receipt for " + username);
        frame.setSize(600, 400);
        frame.setDefaultCloseOperation(JFrame.DISPOSE_ON_CLOSE);

        JPanel panel = new JPanel() {
            private Image backgroundImage = new ImageIcon("C:\\Users\\user\\BlueJ Project\\LabProject\\image100P.jpg").getImage();

            @Override
            protected void paintComponent(Graphics g) {
                super.paintComponent(g);
                // Draw the background image
                g.drawImage(backgroundImage, 0, 0, getWidth(), getHeight(), this);
            }
        };

        panel.setLayout(new BorderLayout(panel, BorderLayout.Y_AXIS));
        panel.setBorder(BorderFactory.createTitledBorder(
            BorderFactory.createLineBorder(Color.GRAY, 2),
            "Receipt",
            TitledBorder.LEADING,
            TitledBorder.TOP,
            new Font("Verdana", Font.BOLD, 16),
            Color.BLACK
        ));
        frame.add(panel);

        // Clear the previous user's data
        datePanel.removeAll(); // Remove all existing buttons from the panel
        receiptArea.setText(""); // Clear the receipt area

        datePanel.setPreferredSize(new Dimension(480, 100));
        datePanel.setBackground(new Color(255, 240, 210));
        datePanel.setLayout(new FlowLayout(FlowLayout.LEFT));

        JScrollPane dateScrollPane = new JScrollPane(datePanel);
        dateScrollPane.setPreferredSize(new Dimension(580, 100));
        panel.add(dateScrollPane, BorderLayout.NORTH);

        receiptArea.setEditable(false);
        receiptArea.setLineWrap(true);
        receiptArea.setWrapStyleWord(true);

        JScrollPane receiptScrollPane = new JScrollPane(receiptArea);
        receiptScrollPane.setPreferredSize(new Dimension(315, 320));
        panel.add(receiptScrollPane, BorderLayout.CENTER);

        JPanel buttonPanel = new JPanel(new BorderLayout());

        JPanel buttonPanel2 = new JPanel(new BorderLayout());
        RoundedButton backButton = new RoundedButton("Main Page", 20);
        buttonPanel.add(backButton, BorderLayout.WEST);
        panel.add(buttonPanel, BorderLayout.SOUTH);

        refreshDatePanel(datePanel, username, receiptArea);

        backButton.addActionListener(new ActionListener() {
            @Override
            public void actionPerformed(ActionEvent e) {
                // Log out and show the login page
                frame.dispose();
                InformationPage.showMainPage(username, datePanel, receiptArea);
            }
        });
        frame.setVisible(true);
    }
}
```

```

// Method to save subscription details to subscription3.txt
protected static void saveSubscriptionDetails(String username, String selectedPackage, double price, String[] selectedSubjects, String[][] selectedTeachers, String date, JPanel datePanel, JTextArea receiptArea) {
    try {
        BufferedWriter writer = new BufferedWriter(new FileWriter("subscription3.txt", true));
        // Write user details with the subscription information
        writer.write("Username: " + username + "\n");
        writer.write("Date: " + date + "\n");
        writer.write("Package: " + selectedPackage + "\n");
        writer.write("Price: RM" + price + "\n");

        // Save the selected teachers for each subject
        for (String[] teacher : selectedTeachers) {
            writer.write("Teacher Name: " + teacher[0] + ", Subject: " + teacher[2] + ", Contact: " + teacher[1] + ", Class: " + teacher[3] + "\n");
        }

        writer.write("-----\n");
        writer.close();
    } catch (IOException e) {
        e.printStackTrace();
        JOptionPane.showMessageDialog(null, "Error saving subscription details. Please try again.", "Error", JOptionPane.ERROR_MESSAGE);
    }
    refreshDatePanel(datePanel, username, receiptArea);
}

// Method to refresh date panel everytime data updated in subscription3.txt and generate new, unique date button
protected static void refreshDatePanel(JPanel datePanel, String username, JTextArea receiptArea) {
    List<String> uniqueDates = new ArrayList<>();
    try (BufferedReader reader = new BufferedReader(new FileReader("subscription3.txt"))) {
        String line;
        boolean isUserSection = false;
        System.out.print("\n*****");
        System.out.print("Checking :...");

        while ((line = reader.readLine()) != null) {
            if (line.startsWith("Username: ")) {
                String currentUser = line.substring(10).trim();
                isUserSection = currentUser.equals(username);

                System.out.println("Found Username: " + currentUser + ", Is User Section: " + isUserSection);
            }

            if (isUserSection && line.startsWith("Date: ")) {
                String date = line.substring(6).trim();

                if (!uniqueDates.contains(date)) {
                    uniqueDates.add(date);
                    System.out.println("Added unique date: " + date);
                }
            }
        }
    } catch (IOException e) {
        System.out.print("Error reading the file for dates.\n");
        e.printStackTrace();
    }

    System.out.println("Unique dates for user: " + uniqueDates);

    datePanel.removeAll();

    // Check for existing buttons to avoid duplicates
    List<String> existingDates = new ArrayList<>();
    for (Component component : datePanel.getComponents()) {
        if (component instanceof JButton) {
            JButton existingButton = (JButton) component;
            existingDates.add(existingButton.getText());
        }
    }

    // Create buttons for each unique date
    for (String date : uniqueDates) {
        if (!existingDates.contains(date)) { // Avoid duplicate buttons
            RoundedButton dateButton = new RoundedButton(date, 20);
            dateButton.addActionListener(e -> displayDataForDate(receiptArea, username, date));
            datePanel.add(dateButton);
            System.out.println("Added new button for date: " + date);
        }
    }

    datePanel.revalidate();
    datePanel.repaint();
    datePanel.revalidate();
    datePanel.repaint();
}

// Method that display data from subscription3.txt everytime date button clicked
private static void displayDataForDate(JTextArea receiptArea, String username, String selectedDate) {
    System.out.println("Displaying data for date: " + selectedDate);
    receiptArea.setText(""); // Clear the text area
    try (BufferedReader reader = new BufferedReader(new FileReader("subscription3.txt"))) {
        String line;
        boolean isUserSection = false;
        boolean isDateSection = false;
        StringBuilder receiptContent = new StringBuilder();
        double totalPrice = 0.0;

        // Read the file line by line
        while ((line = reader.readLine()) != null) {
            System.out.println("Reading line: " + line); // Debug print for each line
            if (line.startsWith("Username: ")) {
                isUserSection = line.substring(10).trim().equals(username);
                System.out.println("Is user section: " + isUserSection);
            }
        }
    }
}

```

```

    if (isUserSection && line.startsWith("Date: ")) {
        String dateFromFile = line.substring(6).trim();
        isDateSection = dateFromFile.equals(selectedDate);
        System.out.println("Is date section: " + isDateSection);
        System.out.println("Added line to receipt: " + line);
    }

    // Process data for the matching user and date
    if (isUserSection && isDateSection) {
        if (line.startsWith("Date: ")) {
            continue; // Skip this line
        }

        receiptContent.append(line).append("\n");

        // Extract and accumulate price
        if (line.startsWith("Price: RM")) {
            String priceString = line.substring(9).trim(); // Remove "Price: RM" and trim spaces
            try {
                totalPrice += Double.parseDouble(priceString);
            } catch (NumberFormatException e) {
                System.out.println("Error parsing price: " + priceString);
            }
        }

        // Stop collecting after the separator line
        if (line.startsWith("-----")) {
            isUserSection = false;
            isDateSection = false;
            System.out.println("Section ended, resetting flags.");
        }
    }
}

if (receiptContent.length() > 0) {
    receiptContent.append("\nTotal Price: RM").append(totalPrice).append("\n");
    receiptArea.append(receiptContent.toString());
} else {
    receiptArea.append("No data found for username: " + username + " on date: " + selectedDate + "\n");
}
System.out.println("Total Price: RM" + totalPrice);
} catch (IOException e) {
    receiptArea.append("Error reading the file for data.\n");
}
}
}

```

5) RoundedButton.java

```

import javax.swing.border.AbstractBorder;
import java.awt.Color;
import java.awt.Component;
import java.awt.Graphics;
import java.awt.Insets;
import javax.swing.JButton;

public class RoundedButton extends JButton {
    private int radius;

    // Constructor to set the button text and the radius for rounded corners
    public RoundedButton(String text, int radius) {
        super(text);
        this.radius = radius;
        setContentAreaFilled(false); // Don't fill the content area to allow custom painting
        setFocusPainted(false); // Remove focus border
        setBorderPainted(false); // Remove border
    }

    @Override
    protected void paintComponent(Graphics g) {
        if (getModel().isPressed()) {
            g.setColor(new Color(255, 99, 71)); // Darker orange when pressed
        } else if (getModel().isRollover()) {
            g.setColor(new Color(255, 140, 0)); // Glitter orange when hovered
        } else {
            g.setColor(new Color(255, 165, 0)); // Soft orange as default
        }

        g.fillRoundRect(0, 0, getWidth(), getHeight(), radius, radius); // Draw rounded rectangle
    }
}

```

```

        super.paintComponent(g); // Paint the button text and other components
    } @Override
    public void paintBorder(Graphics g) {
        g.setColor(new Color(255, 140, 0)); // Border color (orange)
        g.drawRoundRect(0, 0, getWidth() - 1, getHeight() - 1, radius, radius); // Rounded border
    }
}

```

6) InformationPage.java

```

import javax.swing.*;
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;
import java.awt.*;
import java.awt.event.*;
import java.io.*;
import java.util.HashMap;
import java.lang.String;
import javax.swing.border.TitledBorder;

//for main page interface
public class InformationPage{
    protected static void showMainPage(String username, JPanel datePanel, JTextArea receiptArea) {
        JFrame mainFrame = new JFrame("Main Page");
        mainFrame.setSize(600, 400);
        mainFrame.setDefaultCloseOperation(JFrame.DISPOSE_ON_CLOSE);

        // Create a custom panel with a background image
        JPanel panel = new JPanel() {
            private Image backgroundImage = new ImageIcon("C:\\Users\\user\\BlueJ Project\\LabProject\\image100P.jpg").getImage();

            @Override
            protected void paintComponent(Graphics g) {
                super.paintComponent(g);
                // Draw the background image
                g.drawImage(backgroundImage, 0, 0, getWidth(), getHeight(), this);
            }
        };

        panel.setLayout(new BorderLayout(panel, BorderLayout.Y_AXIS));
        panel.setBorder(BorderFactory.createTitledBorder(
            BorderFactory.createLineBorder(Color.GRAY, 2),
            "User",
            TitledBorder.LEADING,
            TitledBorder.TOP,
            new Font("Verdana", Font.BOLD, 16),
            Color.BLACK
        ));

        mainFrame.add(panel);

        // Set frame visibility
        mainFrame.setVisible(true);

        String userData = getUserData(username); // You can fetch full user data like name, email, etc.
        if (userData != null) {
            String[] userDetails = userData.split(",");

            // Display each user detail as a label
            JLabel welcomeLabel = new JLabel("\nWELCOME, " + username + "!");
            welcomeLabel.setForeground(Color.BLACK);
            welcomeLabel.setFont(new Font("Verdana", Font.BOLD, 14));
            panel.add(welcomeLabel);

            JLabel nameLabel = new JLabel("\nName: " + userDetails[1]);
            nameLabel.setForeground(Color.BLACK);
            nameLabel.setFont(new Font("SansSerif", Font.PLAIN, 12));
            panel.add(nameLabel);

            JLabel emailLabel = new JLabel("\nEmail: " + userDetails[2]);
            emailLabel.setForeground(Color.BLACK);

```

```

emailLabel.setFont(new Font("SansSerif", Font.PLAIN, 12));
panel.add(emailLabel);

panel.add(Box.createRigidArea(new Dimension(0, 20)));
}

String[][] coursesAndTeachers = getCoursesAndTeachers(username);
if (coursesAndTeachers.length > 0) {
    // Create a table with columns: Subject, Teacher
    String[] columnNames = {"Subject", "Teacher"};
    JTable table = new JTable(coursesAndTeachers, columnNames) {
        @Override
        public boolean isCellEditable(int row, int column) {
            return false; // Disable editing cells
        }
    };
    table.setFillViewportHeight(true);
    table.setRowHeight(20);
    table.setFont(new Font("Arial", Font.PLAIN, 14));

    // Customize table header
    table.getTableHeader().setFont(new Font("SansSerif", Font.BOLD, 14));
    table.getTableHeader().setBackground(new Color(255, 160, 122)); //light orange header
    table.getTableHeader().setForeground(Color.WHITE);

    //Customize row selection
    table.setSelectionBackground(new Color(255, 179, 89));
    table.setSelectionForeground(Color.BLACK);

    // Alternating row colors
    table.setDefaultRenderer(Object.class, new javax.swing.table.DefaultTableCellRenderer() {
        @Override
        public java.awt.Component getTableCellRendererComponent(JTable table, Object value, boolean isSelected, boolean hasFocus, int row, int column) {
            java.awt.Component c = super.getTableCellRendererComponent(table, value, isSelected, hasFocus, row, column);
            if (row % 2 == 0) {
                c.setBackground(new Color(255, 228, 225)); // light pink for even rows
            } else {
                c.setBackground(new Color(255, 240, 245)); // softer pink for odd rows
            }
            return c;
        }
    });

    JScrollPane scrollPane = new JScrollPane(table);
    scrollPane.setBounds(10, 50, 460, 150); // Attach scroll to the table
    scrollPane.setBorder(BorderFactory.createEmptyBorder());
    panel.add(scrollPane);

    // Add mouse listener to table for detecting clicks on teacher's name
    table.addMouseListener(new MouseAdapter() {
        @Override
        public void mouseClicked(MouseEvent e) {
            int row = table.rowAtPoint(e.getPoint());
            int column = table.columnAtPoint(e.getPoint());
            if (column == 1) { // If Teacher column is clicked
                String teacherName = (String) table.getValueAt(row, 1);
                System.out.println("Clicked Teacher Name: " + teacherName);

                File file = new File("C:\\Users\\User\\BlueJ Project\\First\\Teachers.txt");
                if (!file.exists()) {
                    JOptionPane.showMessageDialog(null, "Teachers file not found!",
                        "Error", JOptionPane.ERROR_MESSAGE);
                }

                return;

                try (BufferedReader reader = new BufferedReader(new FileReader(file))) {
                    String line;
                    boolean found = false;
                    while ((line = reader.readLine()) != null) {
                        System.out.println("Reading line: " + line);
                        String[] details = line.split(",");
                        if ((details.length >= 6) && line.split(",")[0].trim().equalsIgnoreCase(teacherName.trim())) {
                            found = true;
                            String contact = details[1];
                            String subject = details[2];
                            String classGroup = details[3];
                            String imagePath = details[4];
                            String examType = details[5];
                            showTeacherDetailsPopup(teacherName, contact, subject, classGroup, imagePath, examType);
                            break;
                        }
                    }

                    if (!found) {
                        JOptionPane.showMessageDialog(null, "No details found for the selected teacher: " + teacherName,
                            "Error", JOptionPane.ERROR_MESSAGE);
                    }
                } catch (IOException ex) {
                    ex.printStackTrace();
                }
            }
        }
    });
}

```



```

// If there are no courses
if (numEntriesForCourses <= 0) {
    return new String[0][0];
}

// Number of rows will be half of the remaining entries (since courses and teachers are paired)
String[][] tableData = new String[numEntriesForCourses / 2][2];

// Fill the table with course and teacher data
for (int i = 0; i < entries.length; i += 2, i++) {
    String course = entries[i].replace("Registered Course: ", "");
    String teacher = entries[i + 1].replace("Teacher: ", "");
    tableData[i][0] = course;
    tableData[i][1] = teacher;
}

return tableData;
}

} catch (IOException e) {
    e.printStackTrace();
}

return new String[0][0]; // Return empty if no matching data found
}

// Show teacher details in a popup window
private static void showTeacherDetailsPopup(String teacherName, String contact, String subject, String classGroup, String imagePath, String examType) {
    // You can modify this with actual teacher details (from a file, database, etc.)
    String teacherDetails = "Teacher Name: " + teacherName + "\n" +
        "Phone: " + contact +
        "\nSubject: " + subject +
        "\nSubject: " + subject +
        "\nClass: " + classGroup +
        "\nExamType: " + examType +
        "\nProfile Picture: " + imagePath ; // You can replace this with an actual image path

    JOptionPane.showMessageDialog(null, teacherDetails, teacherName + " Details", JOptionPane.INFORMATION_MESSAGE);
}
}

```


CONCLUSION

The Elite Tuition Centre's application offers an efficient solution for the students or parents who face challenges in booking tuition sessions online. This application is user-friendly and gives interactive experience to the users because the system is designed to be simple and understandable. The users can book the schedule appointments smoothly because this application can view available time slots, book sessions and receive the remainder date for each subject subscription.

The application allows users to plan and make decisions whether to continue their sessions. Additionally, users are given the option to choose which teacher that they feel comfortable with to ensure better learning experience. Once the users confirm their choice, the system display the amount of money their have to pay, making the process convenient.

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